Numerical Methods Digest

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A list of references for enhanced retention.

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1. Preliminaries

1.1. Floating-point arithmetic

[Ove01] for detailed explanations.

- 2's complement for signed integers:
  - On 32-bit machine, $x \in [\![0, 2^{31} - 1]\!] \implies$ usual binary representation
  - a negative number $-y$, where $y \in [\![1, 2^{31}]\!]$, is stored as $2^{32} - y$.
  - Arithmetic operations are the same as usual binary operations if overflow digits are discarded.
- Single format:
  - 1 bit for sign, 8 bits for exponent, 23 bits for significand.
  - Range of 2^E: $E_{min} = -126$, $E_{max} = 127$, $E = (a_1 \dots a_8)_2 - 126$.
- Double format:
  - 1 bit for sign, 11 bits for exponent, 52 bits for significand.
  - Range of 2^E: $E_{min} = -1022$, $E_{max} = 1023$, $E = (a_1 \dots a_{11})_2 - 1022$.
- Subtraction of values close together should be avoided when possible due to cancellation.

1.2. Stability and error analysis

[Ove01], [TB97] similar but slightly different notation.
for a problem $f : X \to Y$ defined by $f(x) = y$, and an algorithm $\tilde{f}$ for f,

- Absolute (forward) error of $\tilde{y} = \tilde{f}(x)$: $\|\Delta y\| = \|\tilde{y} - y\|$.
- Absolute backward error: $\inf_{\tilde{x}}\{\|x - \tilde{x}\| : f(\tilde{x}) = y\}$.
- (Forward) stable algorithm $\tilde{f}$:
  - computed answer $\tilde{f}(x)$ is close to $f(\tilde{x})$, i.e. $\|\tilde{f}(x) - f(\tilde{x})\|/\|f(\tilde{x})\| = O(\epsilon)$, where $\|x - \tilde{x}\|/\|x\| = O(\epsilon)$.
  - "Computed answer is nearly the same as an answer to a nearly right question."
- Backward stable algorithm $\tilde{f}$:
  - computed answer $\tilde{f}(x)$ is **equal** to $f(\tilde{x})$, i.e. $\tilde{f}(x) = f(\tilde{x})$, where $\|x - \tilde{x}\|/\|x\| = O(\epsilon)$.
  - "Computed answer is the **same** as an answer to a nearly right question."
- Backward error analysis is multiplying the condition number and backward error to estimate relative forward error.

1.3. Rate of convergence

- Linear convergence:
  - $\exists \beta \in (0,1)$ and integer $K \geq 0$, such that $\forall k \geq K$,

$$|x_{k+1} - x^*| \leq \beta |x_k - x^*|.$$

    If $\beta \to 0$: superlinear, if $\beta \to 1$: sublinear.
- Quadratic/ superlinear convergence:
  - order of convergence is k if there exists finite constant $\beta_k > 0$ such that

$$\lim_{n\to\infty} \frac{|x_{k+1} - x^*|}{|x_k - x^*|^k} = \beta_k.$$

- Richardson extrapolation:
  - For a method $A(h)$ converging to A^* at order n, the order can be improved to $n+1$ by taking $A(h/t)$ and cancelling the biggest term (n^{th} order term) in the truncation error.
  - Namely, take $R(h,t) = \dfrac{t^n A(h/n) - A(h)}{t^n - 1} = A^* + O(h^{n+1})$. (E.g. Romberg's method for integrals in section 6.5)

1.4. Condition number

[Ove01], [TB97] similar but slightly different exposition.

- Absolute condition number:

$$\hat{\kappa}_f = \lim_{\delta\to 0} \sup_{\|\Delta x\| \leq \delta} \frac{\|\Delta f\|}{\|\Delta x\|} = \|J(x)\| \quad \text{(if } f \text{ differentiable)}.$$

- Relative condition number:

$$\kappa_f = \lim_{\delta\to 0} \sup_{\|\Delta x\| \leq \delta} \left(\frac{\|\Delta f\|}{\|f(x)\|]} \Big/ \frac{\|\Delta x\|}{\|x\|} \right) = \lim_{\delta\to 0} \sup_{\|\Delta x\| \leq \delta} \left(\frac{\|\Delta f\|}{\|\Delta x\|} \Big/ \frac{\|f(x)\|}{\|x\|} \right),$$

$$= \frac{\|J(x)\|}{\|f(x)\|/\|x\|} \quad \text{(if } f \text{ differentiable)}$$

- Condition number of Ax w.r.t. perturbations in x:
  - $\kappa = \|A\| \frac{\|x\|}{\|Ax\|} \leq \|A\|\|A^{-1}\|$ (if A invertible).
  - Same as condition number of $A^{-1}b$ w.r.t. perturbations in b.
- Condition number of a matrix $A \iff$ condition number of solving $Ax = b$
  - Sensitivity of x w.r.t. perurbations in A.
  - $\kappa_A = \|A\|\|A^{-1}\| = \dfrac{\sigma_{max}}{\sigma_{min}}$.
- The number of digits the computed solution y agrees with y^*:
  - If an algorithm $\tilde{f}$ is backward stable, for $\tilde{y} = \tilde{f}(x)$ and $y = f(x)$,

$$\text{relative error} = \frac{\|\tilde{y} - y\|}{\|y\|} \leq \kappa_f \frac{\|\tilde{x} - x\|}{\|x\|} = \kappa_f \epsilon, \text{ taking logs of base 10,}$$

$$-\log_{10}\left(\frac{\|\tilde{y} - y\|}{\|y\|}\right) \approx -\log_{10} \kappa_f \epsilon \approx 16 - \log_{10} \kappa_f \quad \text{(using double precision)}.$$

2. Matrix decompositions and identities

2.1. SVD

- Efficient computation: [TB97, Lecture 31], [GV83, p. 8.6.3]. Computing the SVD via eigenvalue problem is unstable. Instead, use two-phase computation:
  - Phase 1: Bidiagonalization ($O(mn^2)$ work).
  - Phase 2: Diagonalize bidiagonal matrix via QR or Divide and conquer ($O(n \log |\log \epsilon|)$ work).
- Low rank approximation: [Mar19, Chapters 3, 4] good reference for overview and randomized methods.
- Pseudoinverse
- Least squares via SVD:

2.2. QR factorization

- Gram-Schmidt:
- Householder transformations:
- Givens rotations:
- Least squares via QR:

2.3. Rank-revealing factorizations

- Eckhart-Young theorem:
- Article by N. Higham

2.4. PLU

2.5. Cholesky

2.6. Schur factorization

2.7. Schur complement

2.8. Hessenberg form

2.9. Polar decomposition

2.10. CUR (Skeleton) decomposition

- Interpolative decomposition
- rank revealing QR

2.11. Nystrom method

2.12. Sherman-Morrison-Woodbury formula

2.13. Sylvester (Weinstein-Aronszajn) determinant identity

3. Methods for eigenvalues

[TB97].

3.1. Power iteration

3.2. Block power iteration (simultaneous iteration)

3.3. Jacobi method for e.vals and e.vecs

3.4. QR algorithm

- Rayleigh quotient
- Inverse iteration

4. Iterative/ Krylov subspace methods

[TB97], [Gen09].

4.1. Non-Hermitian

- Arnoldi iteration
- GMRES
- Biorthogonalization

4.2. Hermitian

- Lanczos iteration
- Conjugate gradient

4.3. Classic iterative methods

- Jacobi
- Gauss-Seidel
- SOR/SSOR

5. Function Approximation

[Tre19], [SM03], [Gen09], [SS03b], [Gra14], [Kre99].

5.1. Gibbs phenomenon

- Fourier series expansion of discontinuous function. E.g., step function, square wave, etc.
- Magnitude of over/undershoot is bounded.
- See exercise 3.20 in [SS03b].
- Numerical experiments with Chebyshev points in [Tre19].

5.2. Runge's phenomenon

- Reminder: Stone-Weierstrass theorem. This only implies the existence of such polynomials. Not any old polynomial.
- Polynomial interpolation error increases at the endpoints of the interval if non-optimal nodes (e.g., equispaced nodes) are used.
- Upper bound of error tends to infinity as n grows.
- Nice examples in [Tre19, §13].

* Fixes:
→ Use Chebyshev points.
→ Piecewise polynomial interpolation 5.7.

5.3. Interpolation formulas

- Lagrange
- Hermite
- Barycentric interpolation formula
- Orthogonal polynomial interpolation
  → Let $V = [\phi_1(\mathbf{x})|\phi_2(\mathbf{x})|\dots|\phi_m(\mathbf{x})] \in \mathbb{R}^{n\times m}$. Namely, the i-th row of V has m orthogonal basis functions evaluated at the i-th entry of $\mathbf{x} \in \mathbb{R}^n$. Also let $\mathbf{f} \in \mathbb{R}^n$ is the vector of tabulated function values of f at the same $\mathbf{x}$, and $c \in \mathbb{R}^m$ the vector of coefficients. Then solving $Vc = \mathbf{f}$ will give us the desired coefficients of the orthogonal polynomial approximation that interpolates f the n points. However, this approach requires solving a system of equations, hence accuracy depends on the conditioning of V. Also, $n \geq m$ is necessary for V to be full rank.
  → Instead, compute approximate coefficients by approximating the inner product integral (in the case of the standard Fourier series):

$$c_i = \frac{1}{L}\int_L f(x)\phi_i(x)dx$$

  of each coefficient with Gauss-Legendre quadrature 6.3. To this end, one must interpolate at n-th order Legendre points, instead of any n points (or interpolate at n-th order Chebyshev points if using Chebyshev-Gauss quadrature instead etc. The gist is to pick the points required for the quadrature formula). Let $M \in \mathbb{R}^{m\times n}$ be the matrix whose i-th row is the i-th basis function evaluated at Legendre points $\mathbf{x} \in \mathbb{R}^n$, multiplied by the quadrature weights, and also the scaling factor of the integral (if necessary). Then a single matrix-vector multiply gives the coefficients of the orthogonal polynomial approximation interpolant, namely, $c = M\mathbf{f}$. Hence, no need to solve a system of equations. Also, no problem even if $n < m$.

5.4. Fourier series

- f periodic ($L = \infty \implies$ Fourier transform), piecewise continuous, piecewise monotonic function.
- Derivation of formulas in e.g., [Kre99].
- Amplitude-phase form $\iff$ cosine-sine form $\iff$ exponential form:
  → See wiki article and Harvard lecture notes by O. Knill.
- f even: use cosine series, f odd: use sine series
- Half-range expansions: f given on $[0, L]$. Instead of finding Fourier series of period L (containing both cosine, sine terms), even or odd extend f to $[-L, L]$ to use cosine or sine series expansion.
- Bessel's inequality, Parseval's identity, L^2 optimality of Fourier series:
  → Nice concise proof and exposition in [WZ15, §8.6].
- Dirichlet conditions: Sufficient conditions for the pointwise convergence of Fourier series. If f satisfies these conditions, then Fourier series converges to midpoint at discontinuity.
  → Conditions: 1. Absolutely integrable over a period, 2. of bounded variation, 3. finite number of discontinuities in any bounded interval, and discontinuities are finite.

$\rightarrow$ Proof that Fourier series converges to the midpoint of jump: Toronto APM346 Notes

- f non-periodic on $\mathbb{R}$, send L to infinity: Fourier series$\rightarrow$ Fourier integral (Inverse Fourier transform). Derivation of the improper Riemann integral in Toronto APM346 Notes.
- Very detailed properties of FT and IFT in $\mathbb{R}$ and $\mathbb{R}^d$ in [SS03b, §5, 6] respectively.
  $\rightarrow$ In $\mathbb{R}^d$,

$$\hat{f}(\xi) = \mathcal{F}(f)(\xi) = \int_{\mathbb{R}^d} f(x)e^{-2\pi i x\cdot\xi}dx, \quad f(x) = \mathcal{F}^*(\hat{f})(x) = \int_{\mathbb{R}^d} f(\xi)e^{2\pi i x\cdot\xi}d\xi.$$

  * FT is a bijective transform on the Schwartz space
  $\rightarrow$ Proof of Fourier Inversion formula, Plancherel formula, Poisson summation formula for functions of rapid decrease (Schwartz class), and extension to functions of moderate decrease in [SS03b]. Proof using complex analysis, and related results in [SS03a].
  $\rightarrow$ Plancherel formula (for f of moderate decrease): $\|\hat{f}\| = \|f\|$.
  Also, for $f, g \in L^2$, $\langle f, g\rangle = \langle \hat{f}, \hat{g}\rangle$, where $\langle\cdot,\cdot\rangle$ is the L^2 inner product.
  $\rightarrow$ Poisson summation formula (for f of moderate decrease):

$$F_1(x) := \sum_{n=-\infty}^{\infty} f(x+n) = F_2(x) := \sum_{n=-\infty}^{\infty} \hat{f}(n)e^{2\pi i n x}.$$

  In particular, when $x = 0$,

$$\sum_{n=-\infty}^{\infty} f(n) = \sum_{n=-\infty}^{\infty} \hat{f}(n).$$

  $F_1(x)$ is called the periodization of f. Note that $F_1(x)$ has period 1. $F_2(x)$ is just the Fourier series of f with period 1. The above formula can be further generalized to arbitrary periods by tinkering with the coefficient in front of n in $F_1(x)$, and scaling the 2π in the exponent of $F_2(x)$, as well as multiplying a scaling factor (the usual Fourier series of period L).
- DFT:
  $\rightarrow$ Can be obtained by fitting N Fourier coefficients using a N-th degree trigonometric polynomial (truncated Fourier series) interpolation using the technique in [Kre99, §11.9].
  $\rightarrow$ DFT can also be retrieved from the Poisson summation formula, which is nicely written out in the wiki article on the Poisson summation formula.
  $\rightarrow$ Other details in [OS99, §8].
- Walsh-Hadamard Transform:
- FFT:
  Quite nice explanation of Cooley-Tukey FFT algorithm in wiki article. For other algorithms: [OS99, §9].
- FWHT:
- Spectral collocation methods: [Tre00], [Tre19, §21], and MIT Notes by S. G. Johnson on spectral differentiation, utilizing the property

$$\mathcal{F}(f')(\xi) = i\xi\mathcal{F}(f)(\xi),$$

  and the fact that FFT, IFFT can be applied in $O(n\log n)$ flops. Plus, advantage over finite differences in accuracy.
  $\rightarrow$ error properties:

5.5. Sturm-Liouville problems and orthogonal polynomials

[Gen09, §4] for overview of different orthogonal systems. [Tre19, §17] for in-depth examples. [Bre11, §5.6] for a thorough treatment of orthonormal bases. [WZ15, §8.6] for Riesz-Fischer theorem and completeness of orthogonal bases.

- Hermite polynomials
- Legendre polynomials
- Chebyshev polynomials

* Cheb and Leg polys are special cases of Jacobi polynomials

5.6. Carathéodory-Fejér approximation

5.7. Piecewise polynomial approximation (splines)

See Exercise 9.8 and notes in Chapter 14 of [Tre19]. Also, [SM03, §11].

- Better suited for discontinuities, and converges exponentially away from the jump (Fourier series converges algebraically away from the jump).
- Also deals with Runge's phenomenon better, ref. 5.2.

6. Numerical Quadrature

[SM03], [Gen09].

6.1. Newton-Cotes formulas

- Midpoint rule
- Trapezoidal rule
- Simpson's rule
- Simpson's 3/8 rule
- Boole's rule
- Composite rules

6.2. Euler-Maclauren expansion

6.3. Gauss and Clenshaw-Curtis quadrature

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- Adaptive quadrature

6.4. Chebyshev-Gauss quadrature

6.5. Extrapolation methods

- Romberg table

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